Exp.no : 17 **RANDOM SAMPLING AND SAMPLING DISTRIBUTION**

AIM:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

PYTHON CODE:

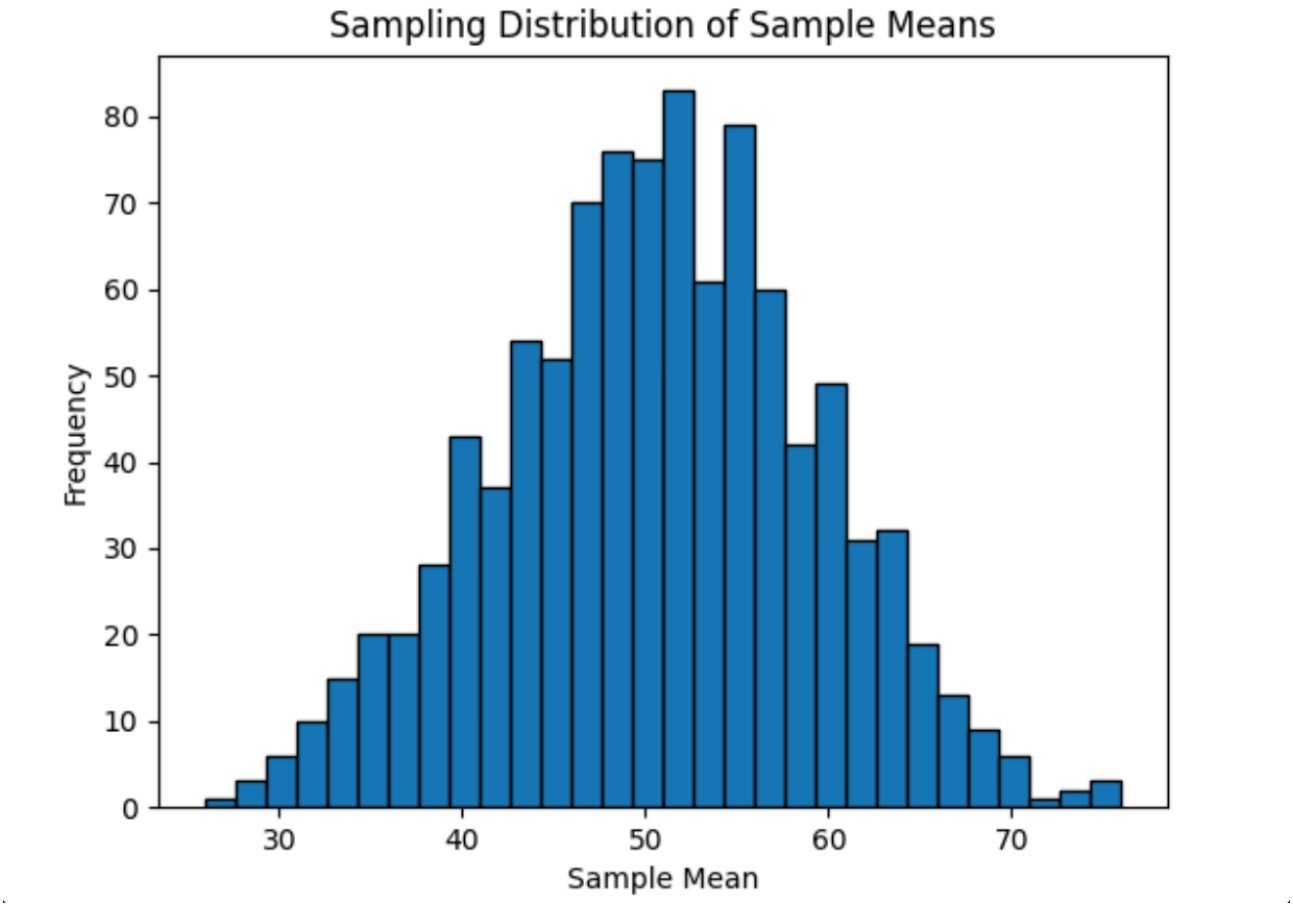
```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
# Load dataset
data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')
# Time Series Decomposition
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
22plt.show()
# Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()
# Anomaly Detection
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
plt.legend()
plt.show()
```

OUTPUT:

```
Population: [  1   2   3   4   5   6   7   8   9  10  11  12  13
14  15  16  17  18
  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34
35  36
  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52
53  54
  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70
71  72
  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88
89  90
  91  92  93  94  95  96  97  98  99 100]Population Mean: 50.5
```

```
Random Sample: [49 97 55 57 92 87 85 90 77 39]
Sample Mean: 72.8

Mean of Sampling Distribution: 50.692899999999995
```



RESULT:

The code is executed successfully.